EV Infra Map India

Challenge 1: Round 1 Submission

Problem understanding | Analytical methodology | Implementation roadmap

Team Information

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Competition ID	EV-F6CC65CBFC96
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Track	Challenge 1: EV Infra Map India

Summary

India has roughly 4.5 million EVs (March 2024) but only about 12,000 public charging stations, a national ratio near 375 EVs per charger against a recommended 20-30. Charging infrastructure is concentrated in a few metros, while most districts, National Highway corridors and Tier-2/3 cities remain underserved. Planners lack a single, trustworthy view of where the gaps are, how they will grow, and where to build next.

EV Infra Map India is an open-data pipeline and web platform that integrates fragmented charger data, fuses it with EV registrations and district boundaries, ranks districts by a transparent **Charging Gap Index (CGI)**, forecasts EV growth under three policy scenarios, estimates charger requirements for 2027 and 2030, and recommends optimal new sites. Every number carries a `data_status` label and every assumption is visible and editable.

1. Problem Understanding

Three pain points sit behind the challenge, and each maps to a component of our solution:

Pain point	Consequence for planners	Our response
Charger data is fragmented across OSM, Open Charge Map, BEE and operators, with duplicates and inconsistent names	No reliable count of what exists	Multi-source ingestion, operator normalisation, spatial + fuzzy deduplication with a confidence score
EV populations and charger locations are never viewed together at district level	Cannot tell which districts are worst-served	VAHAN registrations mapped RTO to district; EVs per charger, coverage and the CGI
No tool to forecast need or choose sites	Investment decided by intuition	Prophet scenarios, requirement formula, MCLP site optimiser

Design principles. (1) Transparency: all assumptions live in one register with a status flag. (2) Honesty about data: observed, derived and synthetic records are labelled, never blended silently. (3) Reproducibility: raw files are stored immutably with timestamps and the pipeline is configuration-driven. (4) Interpretability: planners can see and change why a district ranks where it does.

2. Solution Overview

The platform follows a raw, clean, analytics layering in PostgreSQL/PostGIS, exposed through a FastAPI REST layer and consumed by a React + MapLibre GL front end.

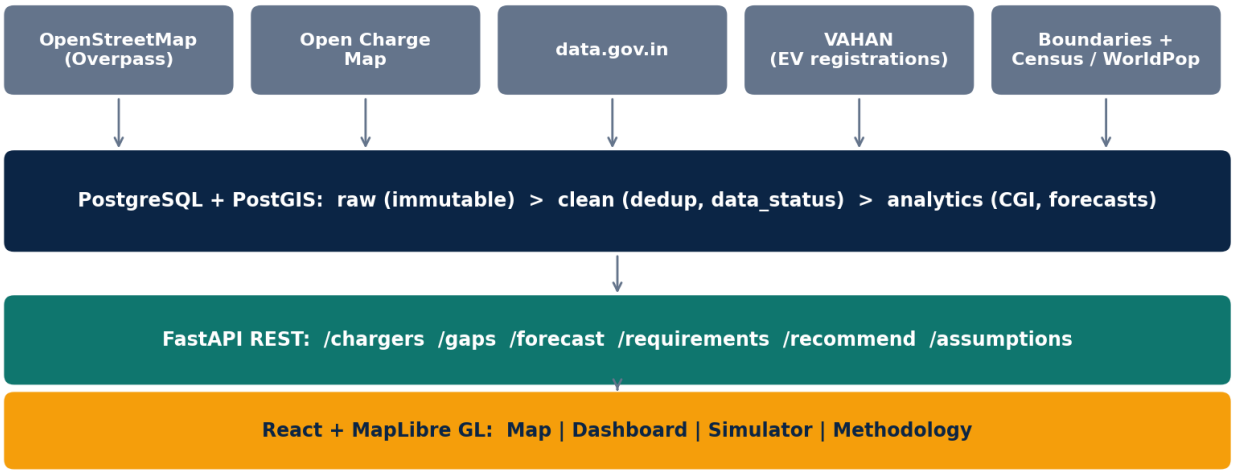


Figure 1: System architecture, from open-data sources to the planner-facing dashboard.

3. The Dashboard and How It Works

The web app has four tabs. The mockup below shows the Map and Dashboard views together; it is an illustration built with placeholder values to explain the layout and interactions, not a capture of real pipeline output.

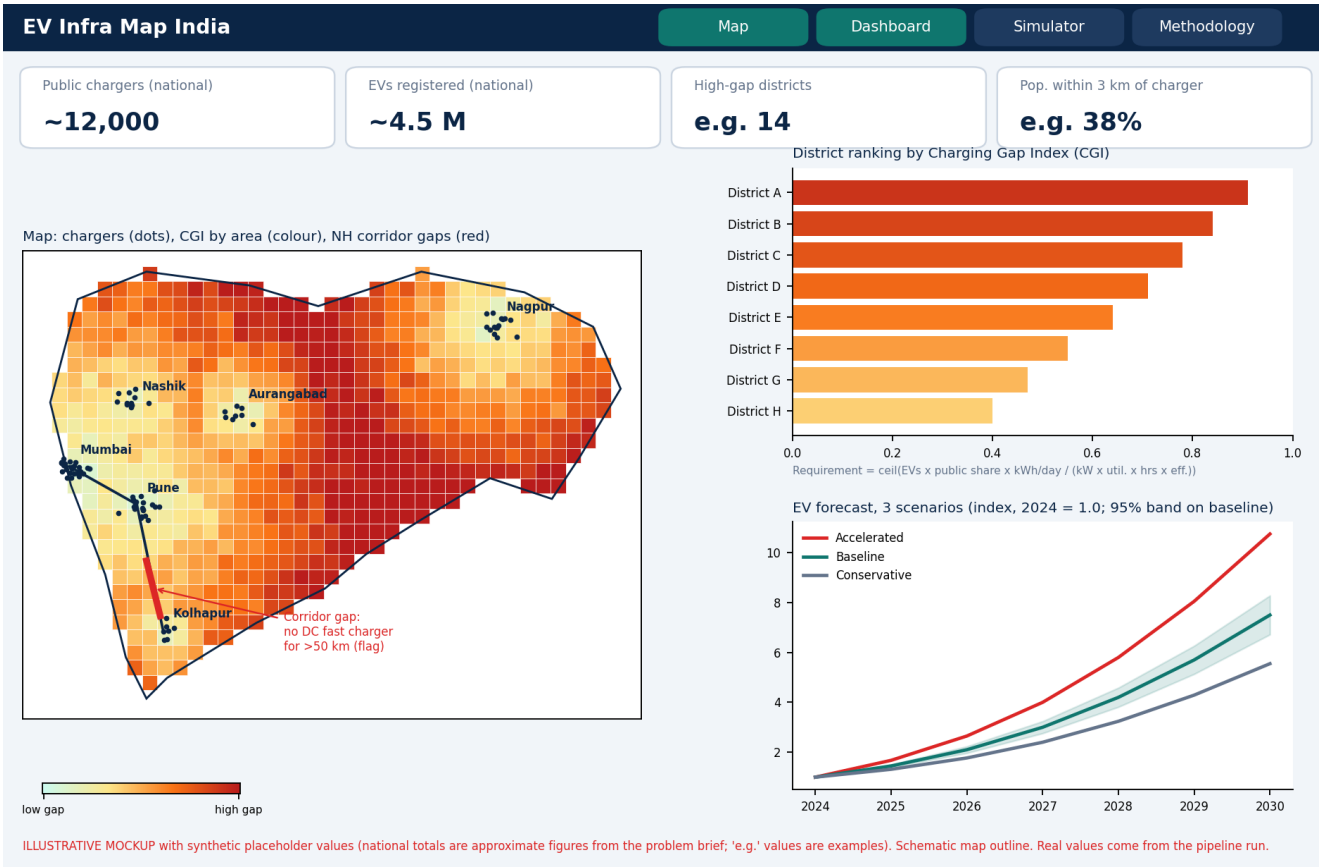


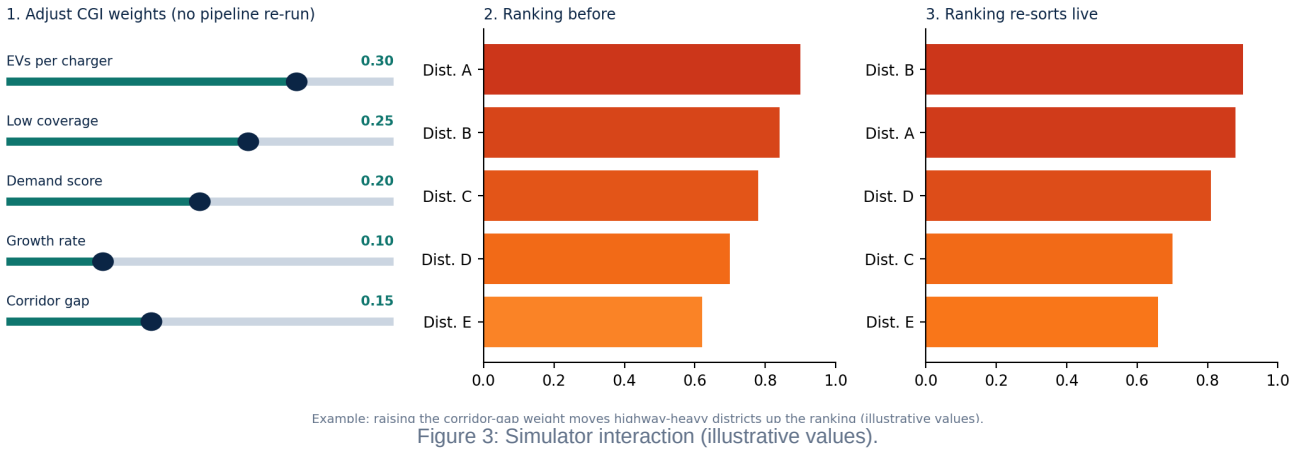
Figure 2: Illustrative dashboard layout (synthetic placeholder values; schematic map outline).

User journey

- **Map tab:** charger points with popups (operator, power, source, confidence score, data-status badge); a gap choropleth coloured by CGI; NH corridor gap overlays; optional coverage buffers.
- **Dashboard tab:** KPI cards, district ranking table, and EV growth charts that switch between baseline, accelerated and conservative scenarios.
- **Simulator tab:** sliders for the five CGI weights; rankings re-sort live without re-running the pipeline.
- **Requirements and recommendations (API):** `GET /requirements` returns shortfall per district, scenario and year; `POST /recommend` returns the top-k MCLP sites with plain-English reasons.
- **Methodology tab:** assumption register, source log and data-status definitions, so every output can be traced.

What-if simulator

Because CGI is a weighted sum of normalised indicators, a planner who prioritises highway access can raise the corridor weight and immediately see which districts move up.



Example: raising the corridor-gap weight moves highway-heavy districts up the ranking (illustrative values).

Figure 3: Simulator interaction (illustrative values).

4. Analytical Methodology

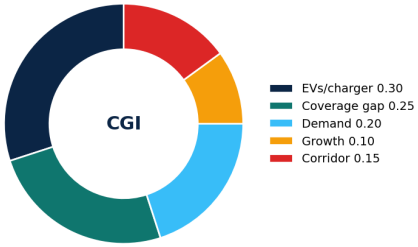
4.1 Charger integration and deduplication

- Sources in priority order: OpenStreetMap (mandatory baseline), Open Charge Map (if API key), data.gov.in (if a usable file exists). Raw records are stored immutably in `raw.chargers_raw`.
- Normalisation: operator alias dictionary plus rapidfuzz (threshold 80); connector types mapped to a canonical set (CCS2, CHAdeMO, Type2, GB/T, Bharat AC-001/DC-001); AC/DC inferred from connector and power.
- Deduplication: DBSCAN on haversine distance (75 m, configurable), then merge within a cluster if operator similarity is at least 80 or distance is under 25 m.
- Each charger is spatially joined to a district with `ST_Within`.

```
confidence = 0.5 x min(n_sources/3, 1) + 0.3 x attribute_completeness + 0.2 x recency
```

4.2 Gap assessment: Charging Gap Index

Indicators per district: total EVs, public and DC-fast chargers, EVs per charger, population within coverage radius, EV growth rate and a demand score. Where a district has zero chargers, EVs per charger is set to the EV count rather than infinity. Indicators are min-max normalised across districts.




```
CGI = 0.30 x norm(EVs per charger) + 0.25 x (1 -
norm(coverage)) + 0.20 x norm(demand) + 0.10 x norm(growth) +
0.15 x norm(corridor gap)
```

Default weights are analyst choices, adjustable in the UI. A rank-stability test perturbs each weight by +/-25% and reports Spearman correlation of district ranks.

Corridor gaps: DC-fast chargers within 1 km of each NH segment are projected onto the road (`ST_LineLocatePoint`); segments whose largest gap exceeds 50 km are flagged. The 50 km threshold is an unverified analyst assumption pending the MoP guidelines.

4.3 Forecasting

- Prophet with yearly seasonality per district and vehicle class; if under 24 months of data exist, a state-level model scaled by district share is used.
- Scenarios as multipliers on the baseline trend: baseline x1.0, accelerated x1.5 (policy incentives such as PM E-DRIVE), conservative x0.7. 95% prediction intervals.
- Backtest: train to a 12-month cutoff, forecast 12 months, report MAPE per district and class.

4.4 Charger requirement estimation

```
daily_energy = forecast_EVs x public_charging_share x kWh_per_EV_per_day
energy_per_charger = charger_kW x utilisation x operating_hours x efficiency
required = ceil(daily_energy / energy_per_charger) | shortfall = max(0, required - existing)
```

Computed per district, vehicle class (2W, 3W, 4W, bus), charger type, scenario and horizon year (2027, 2030). Utilisation is varied by +/-25% in a sensitivity table. Default parameters include a 15% utilisation target, 7.4 kW AC and 30 kW DC chargers, 20 operating hours and 90% efficiency; public-charging shares range from 10% (2W) to 40% (3W).

4.5 Site optimisation

A Maximal Covering Location Problem selects k sites from candidate OSM locations (fuel stations, malls, parking, highway services, excluding sites within 500 m of an existing DC charger) to maximise population-weighted coverage within radius R. It is solved exactly with PuLP + CBC under a 25 s time limit, so results are optimal for the model and fully explainable.

4.6 Validation

- **Deduplication:** before/after counts, cluster-size distribution, and a 100-pair labelled sample for manual review (a human step).
- **Rank stability:** Spearman correlation under +/-25% weight changes.
- **Forecast:** 12-month holdout MAPE.

5. Data Strategy and Tools

Source	Access	Licence	Fallback
OpenStreetMap (Overpass)	REST API, rate-limited	ODbL	Geofabrik extract
Open Charge Map	Public API key	CC BY-SA	OSM-only
VAHAN (MoRTH)	Manual CSV export	Govt. open data	Synthetic allocation by population share (labelled)
DataMeet boundaries	GitHub GeoJSON	ODbL	GADM Level 2
Census 2011	Embedded constants	Public domain	WorldPop raster
WorldPop raster	HTTP download	CC BY 4.0	Census 2011 totals

Quality controls: immutable timestamped raw files; a `data_status` field (observed, derived, synthetic) on every record; dedup confidence scores; automated sanity checks; and an open-questions log for unresolved blockers.

Technology stack: PostgreSQL + PostGIS, Python (pandas, geopandas, scikit-learn, rapidfuzz, osmnx, Prophet, PuLP, scipy), FastAPI, React, Vite, MapLibre GL, Docker Compose.

6. Implementation Roadmap

Phases 0 to 12 (scaffolding through frontend) are built. Remaining effort is documentation and evidence capture.

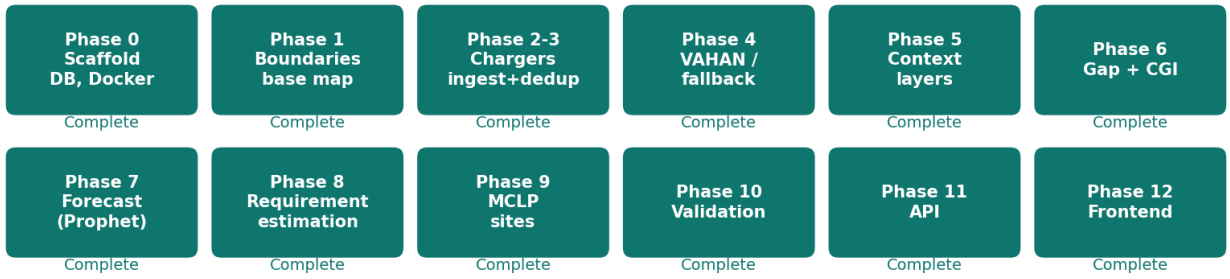


Figure 4: Development stages leading to the final prototype.

Stage	Scope	Tools
0-1	Database schema, Docker, boundaries, base map	PostGIS, FastAPI, Vite, MapLibre
2-5	Charger ingestion and dedup, VAHAN data or fallback, context layers	OSM, OCM, rapidfuzz, DBSCAN, osmnx
6-10	CGI, forecasting, requirements, MCLP, validation	geopandas, Prophet, PuLP, scipy
11-12	REST API and dashboard polish	FastAPI, React, MapLibre GL

7. Risks, Assumptions and Limitations

Risk	Mitigation
VAHAN data not publicly downloadable	Manual export guide; synthetic fallback with clear labelling
Overpass API unavailable	Locally cached Geofabrik extract
No OCM key in time	OSM-only operation; gap noted in source log
District boundaries changed after 2011	2011-vintage boundaries disclosed; open question logged
Policy thresholds not verified	All UNVERIFIED items (DC-fast minimum 30 kW, 3 km urban and 25 km highway radii, 50 km corridor gap) flagged in code and docs

Known limitations. All 22 model parameters in the register are currently analyst assumptions awaiting sourced replacement; OSM charger data is user-contributed and may have gaps; the 1 km corridor proximity is a simplification; Prophet does not model structural breaks such as new policies; VAHAN-based figures are synthetic until a real export is loaded. These are stated openly because the platform is designed to make uncertainty visible rather than hide it.

8. Path to Scale

The pipeline is parameterised by state, so extending beyond the pilot state (Maharashtra) is a configuration change. Planned steps:

- **Jan 2027, 5 states:** state-partitioned tables, vector tiles (pg_tileserv or Martin), scheduled ingestion (Airflow or Prefect).
- **Jun 2027, 15 states:** VAHAN API, BEE EVSE registry feed, DISCOM substation capacity for grid-aware siting.
- **Dec 2027, all states:** live CPO status feeds (OCPI), hierarchical national-state-district forecasting.
- **2028 onward:** role-based access for MHI, ICAT, state nodal agencies and city bodies; automated district planning briefs; full audit trail of assumption changes.

Intended users: MHI, ICAT, state governments and city planners, working from one open, transparent and reproducible platform.